

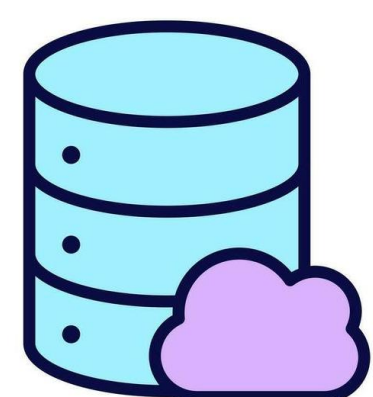
Introduction: Structural elucidation is one of the main challenges in chemistry[1], as it relies primarily on the spectroscopist's expertise. However, the training of these professionals is a lifelong process, which raises the question:

Can machine Learning algorithms predict molecular structures based on infrared (IR) spectroscopic data and their molecular formula?

We present ARCO (*Algorithms of Recognition of Chemical Objects*), a computer-assisted structure elucidation (CASE) system that integrates molecular representations and machine learning algorithms to predict and reconstruct chemical structures from IR spectra.

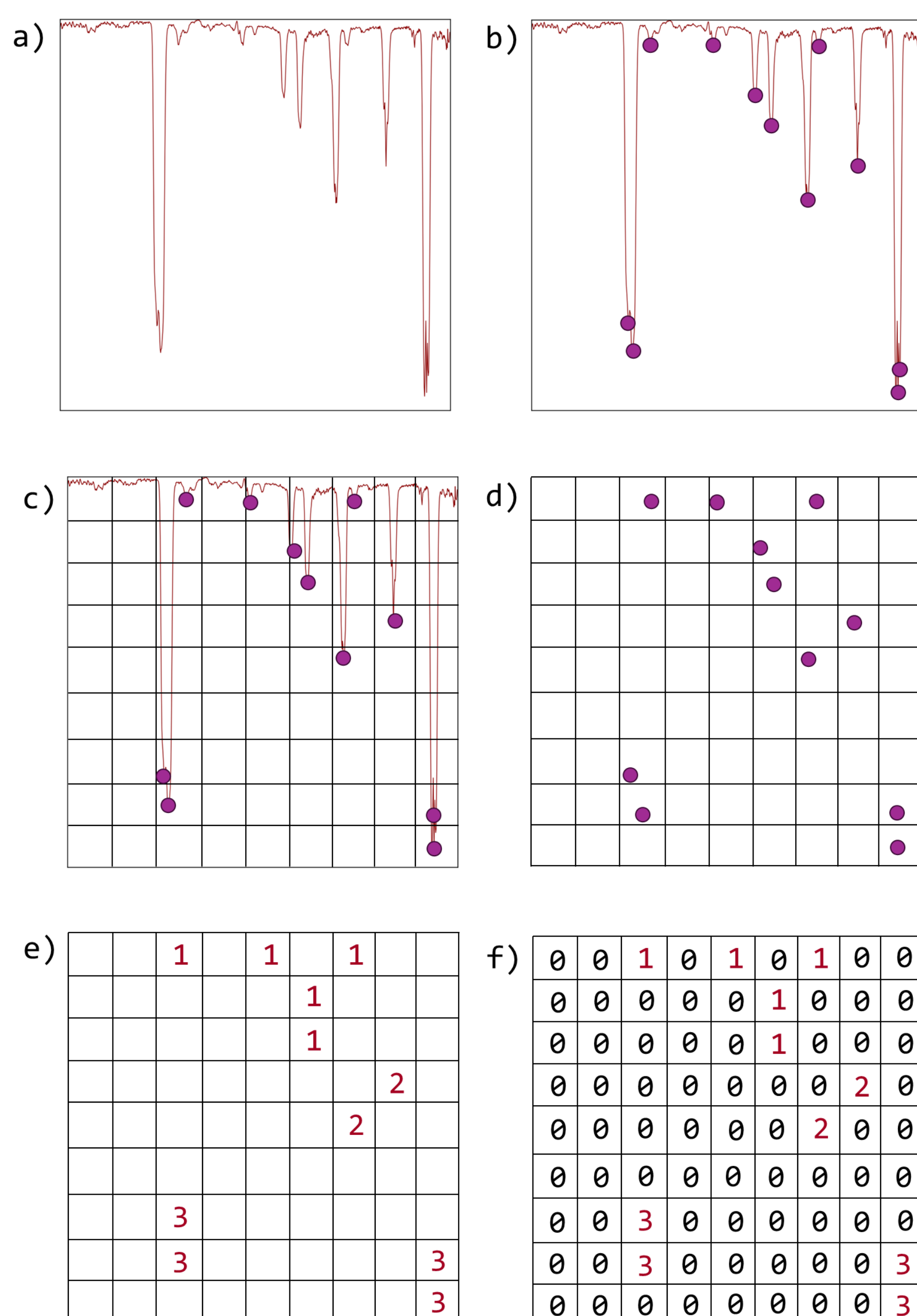
1) Database:

A dataset of 9,875 IR spectra from the NIST (C, H, O, N). Data were split into stratified training (80%) and test (20%).



2) Representation:

- **IR spectrum** → quaternary vector (intensity per cell)
- **Molecule** → TokensSMILES[2] (bonds, not just functional groups)



g) $v_{IR} = [0, 0, 1, 0, 1, \dots, 0, 0, 0, 0, 3]$

3) Models:

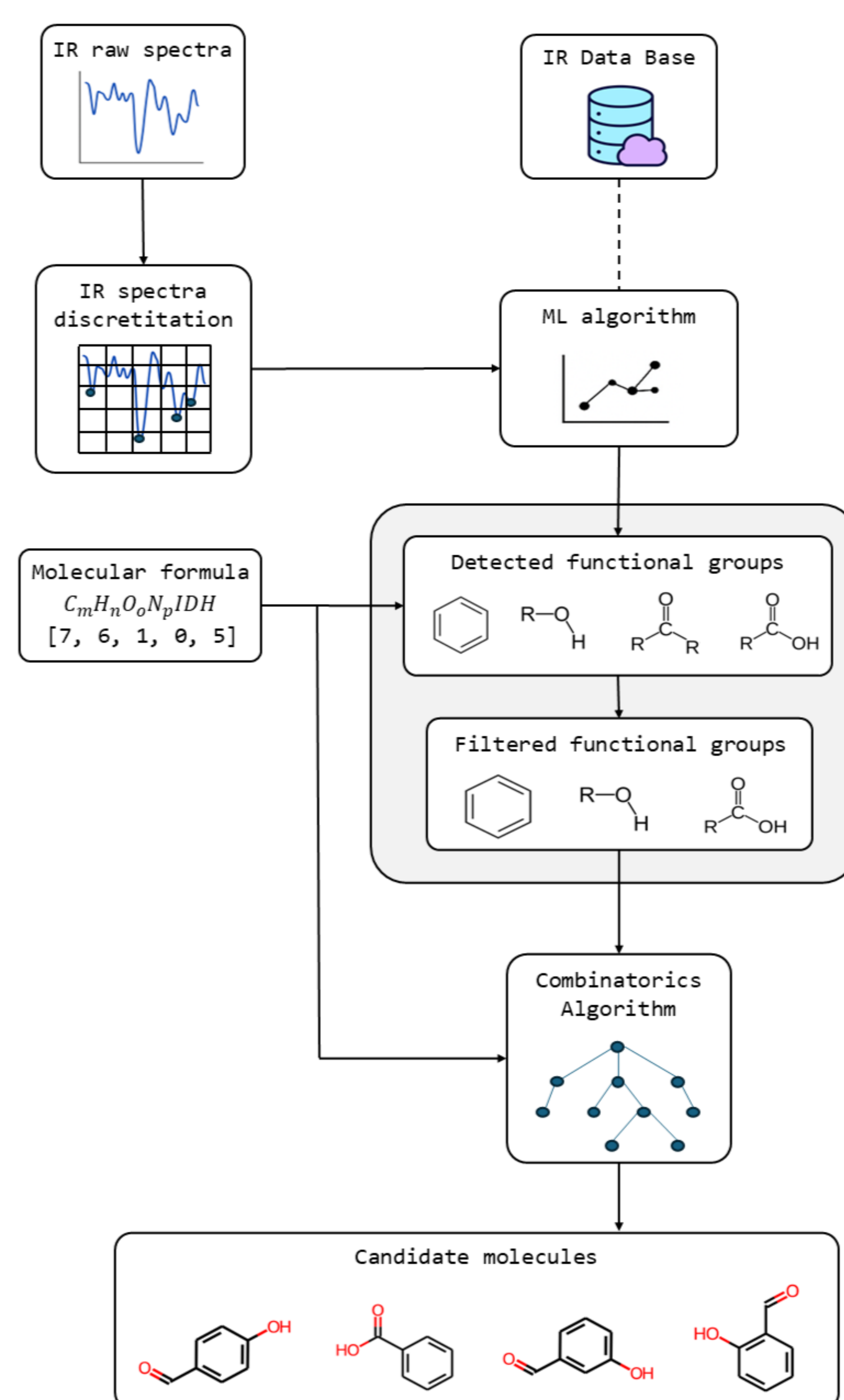
- k-Nearest Neighbors (kNN)
- Support Vector Machine (SVM)
- Random Forest (RF)

Evaluation metric: Macro F1-score

4) Methods:

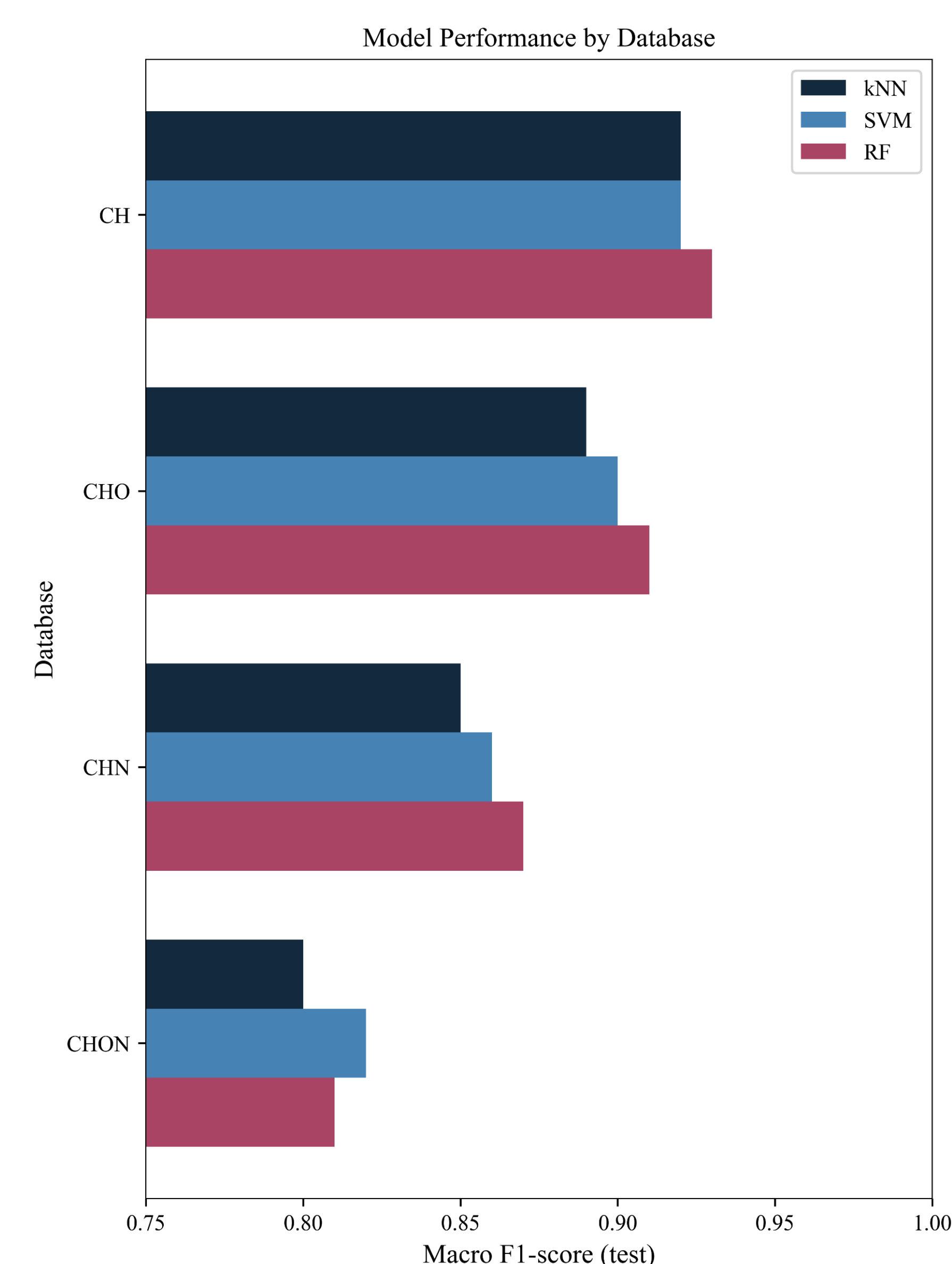
Filter: Molecular formula + chemical constraints.

Generation: Rule-based combinatorial algorithm → Chemically valid structures

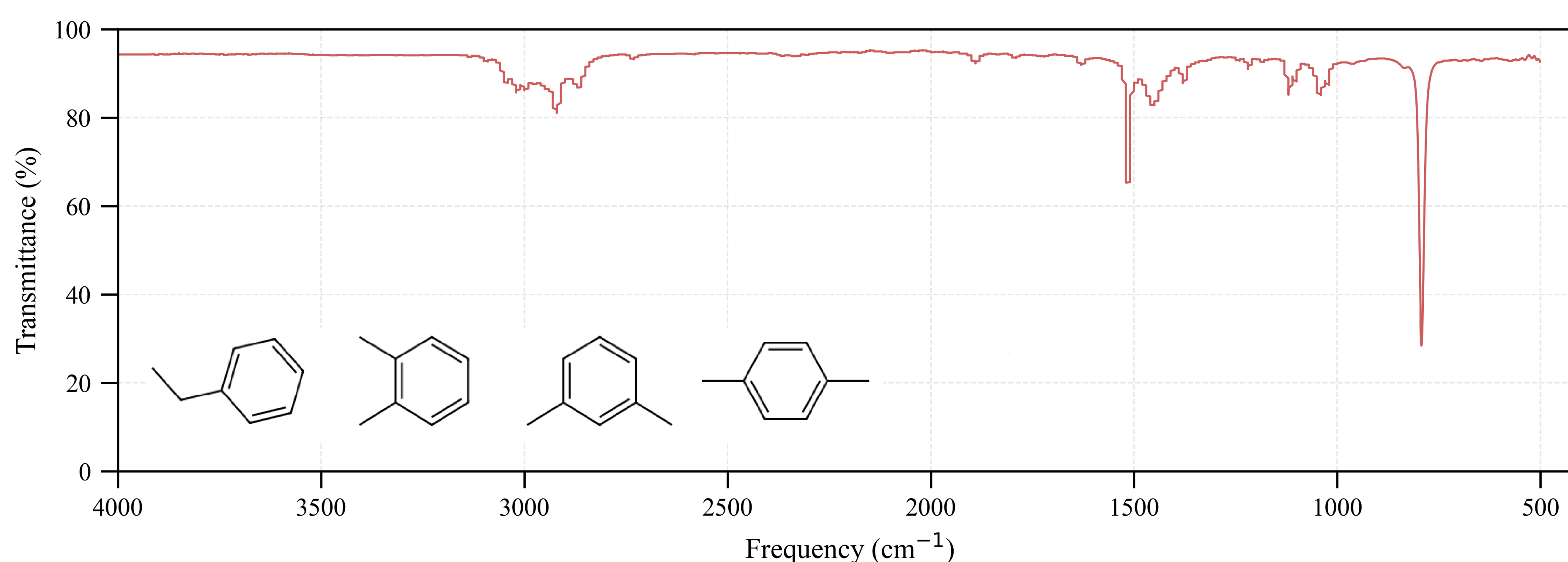


5) Results:

The models achieved macro-F1 scores ranging from 0.80 to 0.93 for bond prediction



The framework was validated under real experimental conditions using the measured IR spectrum of *p*-xylene, successfully identifying the correct structure within the generated candidate space.



Conclusions: ARCO shows that the combination of IR spectra, chemical constraints, and hybrid ML enables robust (F1 0.93) and scalable structural elucidation. As future work, we will implement more spectroscopic techniques, such as NMR or MS.

References:

- (1) Alberts, M.; Hartramp, N.; Laino, T. Automated Structure Elucidation at Human-Level Accuracy via a Multimodal Multitask Language Model.
- (2) Gonzalez-Ortiz, L. A.; Noriega, L.; Ortiz-Chi, F.; Vidales-Ayala, G.; Soberanis-Cáceres, E.; Meneses-Viveros, A.; Aspuru-Guzik, A.; Merino, G. Grammar-Driven SMILES Standardization with TokensSMILES. Chem. Sci. 2026, 17 (3), 1666-1675. <https://doi.org/10.1039/D5SC05004A>.

Acknowledgments:

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